

Aim:

Write a Python program to generate the marksheet of a student. The program should accept the student's name, department, year, semester, section, and roll number, along with the marks obtained in 5 subjects. It should calculate the total marks, percentage, and grade, and display the marksheet in a formatted manner.

Assign grades using the following criteria:

Percentage Grade

90 and above A+

80 to 89.99 A

70 to 79.99 B

60 to 69.99 C

50 to 59.99 D

Below 50 F

Input Format:

The program accepts the following inputs, each entered on a new line:

- Student name, department, year, semester, section, and roll number as strings.
- Marks for 5 subjects as floating-point numbers.

The program displays the following input prompts:

```
Enter Student Name: <name>
Enter Department: <department>
Enter Year: <year>
Enter Semester: <semester>
Enter Section: <section>
Enter Roll Number: <roll_number>
Enter marks for Subject 1: <marks>
Enter marks for Subject 2: <marks>
Enter marks for Subject 3: <marks>
Enter marks for Subject 4: <marks>
Enter marks for Subject 5: <marks>
```

Output Format:

- The program displays a formatted marksheet containing the student's details, the marks obtained in each subject, the total marks, the percentage (formatted to 2 decimal places), and the corresponding grade.
- The output follows this format:

STUDENT MARKSHEET
Name: <name>
Department: <department>
Year: <year>
Semester: <semester>
Section: <section>
Roll No: <roll_number>
Marks Obtained:
Subject 1: <marks>
Subject 2: <marks>
Subject 3: <marks>
Subject 4: <marks>
Subject 5: <marks>
Total Marks: <total_marks>
Percentage: <percentage>
Grade: <grade>

Note: Refer to visible testcases for better clarity.

Source Code:

studentMarksheetGrade.py

```
# Type Content here...
name=input("Enter Student Name: ")
department=input("Enter Department: ")
year=input("Enter Year: ")
semester=input("Enter Semester: ")
section=input("Enter Section: ")
roll_number=input("Enter Roll Number: ")
marks=[]
for i in range(1,6):
    mark=float(input(f"Enter marks for Subject {i}: "))
    marks.append(mark)
total_marks=sum(marks)
percentage=total_marks/5
if percentage>=90:
    grade="A+"
elif percentage>=80:
    grade="A"
elif percentage>=70:
    grade="B"
elif percentage>=60:
    grade="C"
elif percentage>=50:
    grade="D"
else:
    grade="F"

print("STUDENT MARKSHEET")
print(f"Name: {name}")
print(f"Department: {department}")
print(f"Year: {year}")
print(f"Semester: {semester}")
print(f"Section: {section}")
```

```

print(f"Roll No: {roll_number}")
print("Marks Obtained:")
for i in range(5):
    print(f"Subject {i+1}: {marks[i]}")
print(f"Total Marks: {total_marks}")
print(f"Percentage: {percentage:.2f}")
print(f"Grade: {grade}")

```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
Enter Student Name:
Rahul Sharma
Enter Department:
Computer Science
Enter Year:
2
Enter Semester:
4
Enter Section:
A
Enter Roll Number:
101
Enter marks for Subject 1:
85.0
Enter marks for Subject 2:
90.0
Enter marks for Subject 3:
78.0
Enter marks for Subject 4:
88.0
Enter marks for Subject 5:
92.0
STUDENT MARKSHEET
Name: Rahul Sharma
Department: Computer Science
Year: 2
Semester: 4
Section: A
Roll No: 101
Marks Obtained:
Subject 1: 85.0
Subject 2: 90.0
Subject 3: 78.0
Subject 4: 88.0
Subject 5: 92.0
Total Marks: 433.0
Percentage: 86.60
Grade: A

Test Case - 2
User Output
Enter Student Name:
Dhruv
Enter Department:
MME
Enter Year:
1
Enter Semester:
2
Enter Section:
A
Enter Roll Number:
196
Enter marks for Subject 1:
90.0
Enter marks for Subject 2:
90.0
Enter marks for Subject 3:
90.0
Enter marks for Subject 4:
90.0
Enter marks for Subject 5:
90.0
STUDENT MARKSHEET
Name: Dhruv
Department: MME
Year: 1
Semester: 2
Section: A
Roll No: 196
Marks Obtained:
Subject 1: 90.0
Subject 2: 90.0
Subject 3: 90.0
Subject 4: 90.0
Subject 5: 90.0
Total Marks: 450.0
Percentage: 90.00
Grade: A+

Test Case - 3
User Output
Enter Student Name:
Alice
Enter Department:
CSE
Enter Year:
2
Enter Semester:
2
Enter Section:

C
Enter Roll Number:
231
Enter marks for Subject 1:
80.0
Enter marks for Subject 2:
80.0
Enter marks for Subject 3:
80.0
Enter marks for Subject 4:
80.0
Enter marks for Subject 5:
80.0
STUDENT MARKSHEET
Name: Alice
Department: CSE
Year: 2
Semester: 2
Section: C
Roll No: 231
Marks Obtained:
Subject 1: 80.0
Subject 2: 80.0
Subject 3: 80.0
Subject 4: 80.0
Subject 5: 80.0
Total Marks: 400.0
Percentage: 80.00
Grade: A

Test Case - 4
User Output
Enter Student Name:
Bob
Enter Department:
MME
Enter Year:
3
Enter Semester:
1
Enter Section:
B
Enter Roll Number:
215
Enter marks for Subject 1:
70.0
Enter marks for Subject 2:
70.0
Enter marks for Subject 3:
70.0
Enter marks for Subject 4:
70.0

Enter marks for Subject 5:
70.0
STUDENT MARKSHEET
Name: Bob
Department: MME
Year: 3
Semester: 1
Section: B
Roll No: 215
Marks Obtained:
Subject 1: 70.0
Subject 2: 70.0
Subject 3: 70.0
Subject 4: 70.0
Subject 5: 70.0
Total Marks: 350.0
Percentage: 70.00
Grade: B

Test Case - 5
User Output
Enter Student Name:
Charlie
Enter Department:
CSE
Enter Year:
4
Enter Semester:
1
Enter Section:
D
Enter Roll Number:
60.0
Enter marks for Subject 1:
60.0
Enter marks for Subject 2:
60.0
Enter marks for Subject 3:
60.0
Enter marks for Subject 4:
60.0
Enter marks for Subject 5:
60.0
STUDENT MARKSHEET
Name: Charlie
Department: CSE
Year: 4
Semester: 1
Section: D
Roll No: 60.0
Marks Obtained:
Subject 1: 60.0

Subject 2: 60.0
Subject 3: 60.0
Subject 4: 60.0
Subject 5: 60.0
Total Marks: 300.0
Percentage: 60.00
Grade: C